



## TRAINING EXAMPLE — NOT A REAL IEC 62304 DELIVERABLE

This project is a **training site, not a real medical device** under any regulatory definition — it is not Software as a Medical Device (SaMD) or Software in a Medical Device (SiMD). This page is not a genuine IEC 62304 deliverable: it illustrates the *kind* of document a real SaMD/SiMD project would produce. See the [document register](#) for the full explanation.

**Clause:** 5.2 — Software Requirements Analysis · **Register:** [Document Register](#)

### What the standard requires (Class C — every sub-clause of 5.2)

| REF   | TITLE                                                     | APPLIES AT  | OUTPUT                                                                                                                                                                  |
|-------|-----------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.2.1 | Define and document requirements from system requirements | A, B, C     | Documented software system requirements, derived from system-level requirements                                                                                         |
| 5.2.2 | Software requirements content                             | A, B, C     | Functional/capability, inputs/outputs, interfaces, alarms/messages, security, UI, data/database, installation/acceptance, operation/maintenance, IT-network, regulatory |
| 5.2.3 | Include risk control measures in requirements             | <b>B, C</b> | Risk control measures implemented in software, included in the requirements                                                                                             |
| 5.2.4 | Re-evaluate medical device risk analysis                  | A, B, C     | Re-evaluated and updated medical device risk analysis                                                                                                                   |

| REF   | TITLE                        | APPLIES AT | OUTPUT                                                                                                                                         |
|-------|------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.2.5 | Update requirements          | A, B, C    | Re-evaluated and updated requirements, including system requirements                                                                           |
| 5.2.6 | Verify software requirements | A, B, C    | Documented verification that requirements implement system requirements, don't contradict, avoid ambiguity, are testable, unique and traceable |

## 5.2.1 — Where these requirements come from

There is no separate "system" here (see [02 §5.1.3](#)), so the nearest equivalent to a system requirements specification is: **the content of the standard itself**. IEC 62304's own text is the source; this project's requirements are "present that content correctly, accessibly, and interactively," derived and reviewed the way [11](#) describes.

## 5.2.2 — Requirements, by page

Each requirement below is written the way the standard's own content list at 5.2.2 asks for — functional, interface, or otherwise — and points at the `tests/test_site.py` group that verifies it, which is the verification method for 5.2.6.

### Cross-page

| ID     | REQUIREMENT                                                                                                                                      | VERIFIED BY (TEST GROUP) |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| REQ-01 | The site states which edition of IEC 62304 it covers, on every page, and states that Edition 2 is not covered                                    | version                  |
| REQ-02 | Visitors can switch between light and dark colour themes; the choice persists across visits and is not overridden by a later system-theme change | theme                    |

| ID     | REQUIREMENT                                                                                                  | VERIFIED BY<br>(TEST GROUP) |
|--------|--------------------------------------------------------------------------------------------------------------|-----------------------------|
| REQ-03 | No page flashes the wrong theme on load                                                                      | theme                       |
| REQ-04 | All five pages meet WCAG 2.1 A/AA and axe-core best-practice rules, in both themes                           | a11y                        |
| REQ-05 | All five pages remain usable with no horizontal overflow at 1280/768/480/360px and at a 200% zoom equivalent | responsive                  |
| REQ-06 | Every page links to the privacy notice from its footer                                                       | privacy                     |

### Home ( [index.html](#) )

| ID     | REQUIREMENT                                                                                              | VERIFIED BY                                      |
|--------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| REQ-07 | States the standard's edition and that Edition 2 is not covered, in the hero content                     | version                                          |
| REQ-08 | Shows the topic count sourced from the same data the Learn page renders, not a separate hardcoded figure | privacy (home page topic count matches the data) |

### Learn ( [learn.html](#) )

| ID     | REQUIREMENT                                                                                                                                                     | VERIFIED BY |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| REQ-09 | Renders one card per topic from <code>data/phases.json</code> , asynchronously, with loading/error/retry states                                                 | learn       |
| REQ-10 | Each card expands/collapses by mouse and keyboard (Enter/Space), with <code>aria-expanded</code> kept correct                                                   | learn       |
| REQ-11 | A level toggle (Introductory/Advanced) swaps bullet content in place without losing expanded or studied state; the choice persists to <code>localStorage</code> | learn       |

| ID      | REQUIREMENT                                                                                                                                                                                         | VERIFIED BY   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| REQ-12  | A safety-class filter (A/B/C/all) shows/hides cards using the sub-clause data, not a hardcoded per-clause list, and flags partially-applicable cards                                                | applicability |
| REQ-13  | Selecting a class reveals a live-region notice naming every omitted or partially-applicable area and why, including a standing ISO 14971 caution                                                    | learn         |
| REQ-14  | Selecting a class generates a deliverables list (on screen, CSV, and print) of the documented outputs required at that class                                                                        | deliverables  |
| REQ-15  | The site refuses to render Learn content if <code>phases.json</code> and <code>applicability.json</code> disagree on a clause's classes, and names the disagreement                                 | applicability |
| REQ-16  | A progress tracker lets a visitor mark topics as studied and shows a running count                                                                                                                  | learn         |
| REQ-16a | A topic cannot be marked studied until its example document has been opened (Preview or Download); a blocked attempt shows an inline message, expands the card, and moves focus to the Preview link | learn         |

#### Quiz ( [quiz.html](#) )

| ID     | REQUIREMENT                                                                                                       | VERIFIED BY |
|--------|-------------------------------------------------------------------------------------------------------------------|-------------|
| REQ-17 | Two 15-question sets (Introductory/Advanced); only the set matching the visitor's chosen level is downloaded      | quiz        |
| REQ-18 | Questions are presented in randomised order with a 30-second per-question timer and immediate feedback            | quiz        |
| REQ-19 | An 80% pass mark produces a certificate reflecting the training level completed, downloadable and print-formatted | quiz        |
| REQ-20 | A timed-out or failed run is handled explicitly, not just a passing run                                           | quiz        |

### Contact ( [contact.html](#) )

| ID     | REQUIREMENT                                                                                                                                   | VERIFIED BY |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| REQ-21 | Client-side validation runs on blur and on submit, per field                                                                                  | contact     |
| REQ-22 | Submission is asynchronous (no page reload), with a "Sending..." state, a request timeout, and field-level error reporting for a 422 response | contact     |
| REQ-23 | A double-submit is prevented while a request is in flight                                                                                     | contact     |
| REQ-24 | A hidden honeypot field deters spam without being visible or reachable by a real visitor                                                      | contact     |

### Privacy ( [privacy.html](#) )

| ID     | REQUIREMENT                                                                                                                              | VERIFIED BY |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| REQ-25 | Names what the contact form collects, the transfer to Formspree (US), the retention period, and how to contact the supervisory authority | privacy     |
| REQ-26 | States the provenance of the course content, the safety-class mapping, and the Edition 2 status, and disclaims any IEC affiliation       | privacy     |

## 5.2.3 — Risk control measures included in the requirements

The one place an actual "risk" exists in this project is content correctness (see [11](#)). The control measure is REQ-15 above — the site's refusal to render on contradictory safety-class data — which is itself a requirement, not just an implementation detail, for exactly that reason.

## 5.2.4–5.2.5 — Re-evaluation

Each of the three logged content defects described in [11 — Risk Management File](#) triggered a re-evaluation of the requirements above (specifically REQ-12 and REQ-15, which did not exist before the defects were found) — this is that process's paper trail.

## 5.2.6 — Requirements verification

Verification method, per requirement, is the test group named in the tables above. This is coarser than the "verify every requirement individually" a formal SRS review would produce — the granularity available is *group*, not *case*, and that is recorded here rather than implied to be finer than it is.

### Gaps

- Requirements are traced to test **groups**, not individual test cases — the check labels inside `tests/test_site.py` are the case-level detail, but there is no separate traceability matrix connecting REQ-IDs to specific `R.check(...)` calls. For a project this size the group-level mapping above is proportionate; it would not be for a larger one.
- This document was written **after** the features it describes, not before them — for a real Class C project 5.2.1 expects requirements to exist before implementation. That ordering gap is itself an example of the kind of finding [13 — Problem Resolution](#) exists to catch and record, not just a footnote.